Course Outline of CUHK(SZ)
Spring Term 2023

1.Course Identity

The information in this block should be exactly as approved by CUHK Senate.

Course code	STA2002
Course title (English)	Probability and Statistics II
Course title (Chinese)	概率及统计(二)
Units	3
Description (English)	This course is designed to study some advanced concepts of probability and statistics. Topics include: Descriptive measures and regression. Estimation, maximum likelihood and Bayesian. Interval estimation. Estimation in linear regression models. Inference: hypothesis testing, significance and power, P-value, likelihood ratio tests. Analysis of Variance. Distribution free tests and goodness-of-fit tests.

2.Prerequisites / Co-requisites

Please state prerequisites and co-requisites, in terms of courses in CUHK (SZ)* or any other requirements (e.g., having taken certain subjects in high school).

(* Because course codes may not yet be stable, please provide both course code and course tile.)

A.Prerequisites

STA2001: Probability and Statistics I or ECO2121 Basic Statistics

B.Co-requisites

None

3.Learning Outcomes

After taking this course, students are expected to achieve the following outcomes.

Knowledge Outcomes:

- Basic knowledge in descriptive statistics: measures and regression.
- Basic knowledge of confidence intervals, and common hypothesis tests, including t-test, chi-square test, Wilcoxon test
- Basic knowledge of experimental design

Skills Outcomes:

- Applying statistical summary techniques.
- Ability to estimate unknown parameters in different models and testing hypotheses.
- Interpreting the estimate and the results of hypotheses testing and confidence intervals correctly both in statistical terms and in Nayman language.
- Ability to apply regression analysis to conduct data analysis
- Ability to apply ANOVA for data analysis

Value Outcomes:

- Understand the basic philosophy of statistical data analysis.

4.Assessment Scheme

Component/ method	**% weight**
Assignments and Quizzes	**20**
Mid-Term Exam	**25**
Final Exam	**55**

*the number of assignments/quizzes/reports is subject to change as the instructor sees fit.

5.Feedback for Evaluation

- Discussions in office hours;
- Discussions after lecture/tutorial;
- Course Teaching Evaluation (CTE) by the end of the semester.

6.Information of Textbooks

A.Required

- Hogg, R. V., Tanis, E. A. Zimmerman, D. L. (2013) Probability and Statistical Inference, 9th Edition, Pearson.

B.Recommended

- Hogg, McKean and Craig (2018) Introduction to mathematical statistics, 8th Edition, Prentice Hall.

7.Course Components

Teaching Activities	Hours/week
Lecture Session	3 hours per week
Tutorial Session	1 hour per week

8.Tentative Teaching Plan

The teaching progress and sequence may be slightly adjusted along the semester.

Week	Content/ topic/ activity
1	Descriptive statistics. Central measures: arithmetic, geometric and harmonic means. The median.
2	Dispersion measure. Standard unit. Chebyshev's inequality. Histograms.
3	Correlation. Least squares regression line.
4	Review of families of distributions.
5	Generating point estimators for parameters: method of moments, maximum likelihood estimators, Bayesian estimators.
6	Evaluating estimators. Unbiasedness, mean square error, consistency, best linear estimators.
7	Point estimation in a simple linear model.
8	Introduction to interval parameter estimation. **Midterm Exam.**
9	Confidence intervals for the mean, proportion, difference in means, variance, slope and intercept in simple linear model.
10	Inference. Testing hypothesis, tests and test statistics, confidence level and power of tests. P-value.
11	Testing for the mean, proportions, difference in means, variance, slope and intercept in simple linear model.
12	Analysis of variance. Contingency tables. One-way analysis. Testing for homogeneity.
13	Goodness-of-fit tests. Chi-square tests.
14	Non-parametric tests: Permutation test, Wilcoxon test, the sign test.
15	**Final Exam**